KEDGE BUSINESS SCHOOL · QM1 · 2026–2027

Worksheet 3

Dispersion and concentration

Fully worked solutions

Badr ABOUFARASSE, PhD

1. Core applications

Master the essential definitions and calculations from the course.

Core application 25 min

Exercise 4 — Medial and cumulative relative masses

The spending of 40 customers (€) is grouped below. Approximate class monetary mass by $n_i \times$ class midpoint.

Classe / Class	[10,20[	[20,30[	[30,40[	[40,50[	[50,60[
n_i	8	12	10	6	4

1. Compute midpoints, masses, cumulative masses and relative cumulative mass shares.
2. Interpolate the medial in the class containing half of total mass.
3. Compute the frequency median and compare the two statistics.

Exercise 4 — Solution — medial

Classe / Class	n_i	c_i	$n_i c_i$	Masse cumulée	Masse cumulée (%)
[10,20[	8	15	120	120	9.52
[20,30[	12	25	300	420	33.33
[30,40[	10	35	350	770	61.11
[40,50[	6	45	270	1040	82.54
[50,60[	4	55	220	1260	100
TOTAL	40	—	1260	—	100

$$S = \sum n_i c_i = 120 + 300 + 350 + 270 + 220 = 1260, \qquad S/2 = 630.$$

Half the mass lies in [30,40[, since 420 < 630 ≤ 770. With linear interpolation of the mass:

$$MI = 30 + \frac{630 - 420}{350} \times 10 = 36 \text{ euros.}$$

The frequency median is €30 (20 of 40 observations below €30). The medial, €36, divides monetary mass, not the number of observations.

Core application 20 min

Extra 1 — Dispersion from a frequency table

The number of weekly complaints per store is distributed as follows:

Complaints x	**0**	**1**	**2**	**3**	**4**
Stores n	4	8	10	6	2

1. Calculate the mean, population variance and standard deviation.
2. Calculate the coefficient of variation.
3. Interpret every result in context.

Build columns nx and nx². Use E(X²) − [E(X)]² for the population variance.

Extra 1 — Dispersion from a frequency table

Detailed solution

Totals. N = 30; Σnx = 0+8+20+18+8 = 54; Σnx² = 0+8+40+54+32 = 134.

Mean. 54/30 = **1.80 complaints**.

Variance. E(X²) = 134/30 = 4.4667. Therefore σ² = 4.4667 − 1.80² = **1.2267**.

Standard deviation. σ = √1.2267 = **1.108 complaints**.

Coefficient of variation. CV = 1.108/1.80 × 100 = **61.6%**.

Interpretation. Stores receive 1.8 complaints on average, with substantial heterogeneity relative to that mean. The CV is meaningful here because the ratio-scale variable has a genuine zero, but comparisons should use the same observation period and definition of a complaint.

x_i	n_i	$n_i x_i$	$n_i x_i^2$
0	4	0	0
1	8	8	8
2	10	20	40
3	6	18	54
4	2	8	32
Σ	30	54	134

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 35 min

Exercise 2 – Dispersion of grouped data

Using the 210-customer distribution from Worksheet 2, calculate $\sum n_i c_i$, $\sum n_i c_i^2$, mean, population variance, standard deviation, coefficient of variation and IQR. Compare the information provided by standard deviation and IQR.

Exercise 2 – Detailed solution

$$\sum n_i c_i = 6665, \qquad \sum n_i c_i^2 = 227036.5.$$

$$\bar{x} = \frac{6665}{210} \approx 31.7381.$$

$$V = \frac{227036.5}{210} - \bar{x}^2 \approx 73.8195.$$

$$\sigma \approx 8.5918, \qquad CV = \frac{8.5918}{31.7381} \approx 27.07\%.$$

$$IQR = 37.50 - 26.56 \approx 10.94.$$

Standard deviation uses the full distribution and is sensitive to extremes; the IQR describes the spread of the central half and is more robust.

Weighted-product audit table

Class	n	c	n×c	n×c²
[10,14[	4	12	48	576.00
[14,18[	12	16	192	3072.00
[18,26[	32	22	704	15488.00
[26,30[	32	28	896	25088.00
[30,35[	60	32.5	1950	63375.00
[35,40[	35	37.5	1312.5	49218.75
[40,45[	25	42.5	1062.5	45156.25
[45,50[	5	47.5	237.5	11281.25
[50,55[	5	52.5	262.5	13781.25
Σ	210	—	6665	227036.50

$$V = \frac{227036.5}{210} - \left(\frac{6665}{210}\right)^2 = 73.8195, \quad \sigma = \sqrt{73.8195} = 8.5918.$$

Midpoint-based variance approximates the dispersion of the unknown individual observations.

Assessment / exam practice 30 min

Exercise 3 – Lorenz curve and Gini coefficient

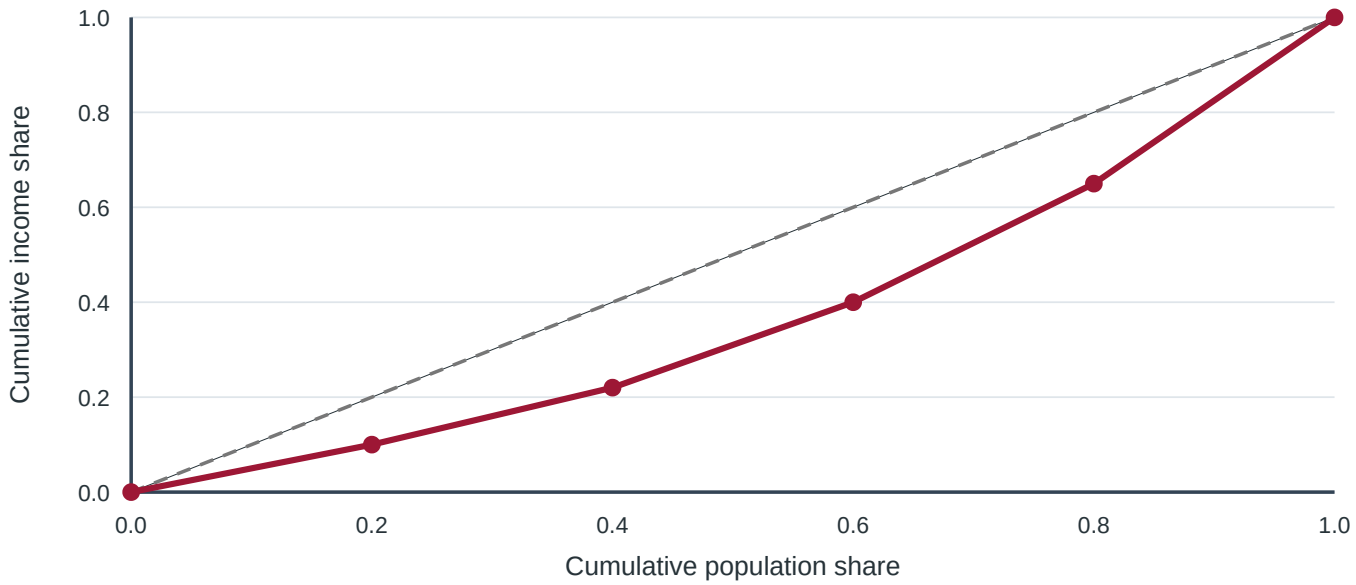
Income values are $10, 12, 18, 25, 35$. Calculate cumulative population and income shares, construct the Lorenz curve, calculate the area by trapezoids, derive the Gini coefficient and explain why Gini measures concentration without explaining its causes.

Exercise 3 – Detailed solution

Total income is 100. Lorenz points are

$$(0,0),\ (0.2,0.10),\ (0.4,0.22),\ (0.6,0.40),\ (0.8,0.65),\ (1,1).$$

$$A = 0.374, \qquad G = 1 - 2A = 0.252 = 25.2\%.$$



Lorenz curve and equality line

The Gini coefficient summarizes concentration; it does not identify the economic or institutional mechanisms causing it.

3. Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice 30 min

Exercise 1 – Boxplot, potential outliers and variance choice

Consider

$$5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 11, 20.$$

Use the median-of-halves convention for quartiles.

1. Calculate the five-number summary.

2. Calculate the range and IQR.
3. Calculate the fences $Q_1 - 1.5IQR$ and $Q_3 + 1.5IQR$ and identify potential outliers.
4. Calculate the mean and mean absolute deviation.
5. Calculate population variance and standard deviation if the 12 observations are the entire population.
6. Calculate sample variance and standard deviation if they are a sample.

Exercise 1 – Detailed solution

$$\min = 5, \qquad Me = 8, \qquad \max = 20.$$

Lower half $5, 6, 6, 7, 7, 8$ gives

$$Q_1 = \frac{6+7}{2} = 6.5.$$

Upper half $8, 9, 9, 10, 11, 20$ gives

$$Q_3 = \frac{9+10}{2} = 9.5.$$

Thus

$$IQR = 9.5 - 6.5 = 3.$$

The fences are

$$B_L = 6.5 - 1.5(3) = 2, \qquad B_U = 9.5 + 1.5(3) = 14.$$

Since $20 > 14$, 20 is a potential outlier.

$$\bar{x} = \frac{106}{12} \approx 8.8333.$$

$$MAD = \frac{1}{12} \sum |x_i - \bar{x}| \approx 2.4722.$$

Population variance and standard deviation:

$$\sigma^2 \approx 14.1389, \qquad \sigma \approx 3.7602.$$

Sample variance and standard deviation:

$$s^2 \approx 15.4242, \qquad s \approx 3.9274.$$

Table of every deviation

i	x	$\lvert x-106/12\rvert$	$(x-106/12)^2$
1	5	3.8333	14.6944
2	6	2.8333	8.0278
3	6	2.8333	8.0278
4	7	1.8333	3.3611
5	7	1.8333	3.3611

i	x	$\lvert x-106/12\rvert$	$(x-106/12)^2$
6	8	0.8333	0.6944
7	8	0.8333	0.6944
8	9	0.1667	0.0278
9	9	0.1667	0.0278
10	10	1.1667	1.3611
11	11	2.1667	4.6944
12	20	11.1667	124.6944
Σ	106	29.6667	169.6667

$$\sum x_i^2 = 1106, \quad \sum (x_i - \bar{x})^2 = 1106 - \frac{106^2}{12} = 169.6667.$$

$$e = 29.6667/12 = 2.4722, \quad \sigma^2 = 169.6667/12 = 14.1389, \quad s^2 = 169.6667/11 = 15.4242.$$

Table entries are rounded for display; use exact fractions until the final rounding.

Advanced practice 25 min

Extra 2 — Comparing concentration and robustness

Two markets have the following ordered firm market shares:

Market A	10%	15%	20%	25%	30%
Market B	2%	8%	10%	20%	60%

1. Construct cumulative shares for each market.
2. Explain which Lorenz curve is farther from equality.
3. For each market calculate the range and discuss why it is insufficient.
4. State a defensible concentration conclusion.

With five equal-weight firms, cumulative population shares increase by 20 percentage points.

Extra 2 — Comparing concentration and robustness

Detailed solution

Cumulative shares. Market A: 0%, 10%, 25%, 45%, 70%, 100%. Market B: 0%, 2%, 10%, 20%, 40%, 100%. Population coordinates are 0%, 20%, 40%, 60%, 80%, 100%.

Lorenz comparison. At every interior population share, B’s cumulative market share is lower. Its Lorenz curve is therefore farther below the equality line: B Lorenz-dominates in the direction of greater concentration.

Ranges. A: 30−10 = 20 points. B: 60−2 = 58 points. The range confirms a difference but uses only two firms and ignores the middle of the distribution.

Conclusion. Market B is clearly more concentrated: the largest firm alone controls 60%, and the bottom 80% of firms jointly control only 40%. A complete report should show the Lorenz curve or a concentration index and should not infer market power from shares alone.